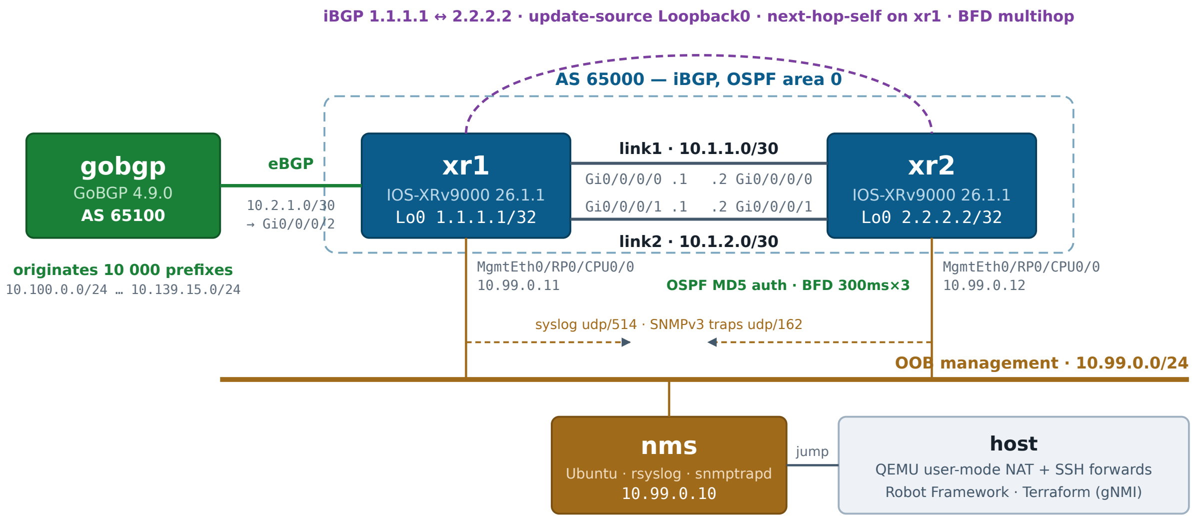
Cisco IOS-XRv9000 Lab — Topology & Testbed

Four QEMU/KVM guests on one host, no root required · OSPF + iBGP core with BFD, external eBGP speaker, out-of-band management · configured with Cisco Network as Code · `github.com/dcantor/IOS-XR-NAC`

> **Lab environment only.** Credentials are committed in plaintext and transport security is disabled throughout, deliberately — see the repository README. Do not reuse either in a production network. Figures here are read from the live lab and its committed configuration.

TOPOLOGY



Solid = data plane. Dashed purple = iBGP session over loopbacks. Dashed orange = syslog and SNMPv3 traps; nms also polls the routers on udp/161 in the other direction. The routers' management ports exist only on the OOB network; the host reaches them through **nms**.

NODES

NODE	ROLE	IMAGE	VCPU	RAM
xr1	Router, eBGP edge	IOS-XRv9000 26.1.1	4	20 GiB
xr2	Router	IOS-XRv9000 26.1.1	4	20 GiB
nms	Syslog + SNMP server, jump host	Ubuntu 24.04 minimal	1	1 GiB
gobgp	External BGP speaker	Ubuntu 24.04 + GoBGP 4.9.0	1	1 GiB

BGP

SESSION	TYPE	LOCAL AS	PEER AS	PREFIXES
`1.1.1.1 ↔ 2.2.2.2`	iBGP	65000	65000	10 000
`10.2.1.1 ↔ 10.2.1.2`	eBGP	65000	**65100**	10 000

iBGP peers loopback-to-loopback (`update-source Loopback0`), so the session rides the OSPF-learned /32 and survives either link failing. xr1 applies `next-hop-self` so xr2 can resolve the external prefixes.

HOST REQUIREMENTS

RESOURCE	NEEDED	NOTE
RAM	**≈ 42 GiB**	20 GiB per router is not negotiable — below ~18 GiB the XRv9k host OS wedges during boot with nothing logged.
Disk	≈ 15 GiB	Copy-on-write overlays; the 2 GB base image is never written to.
Virtualisation	`/dev/kvm`	Nested virtualisation required — XRv9k runs XR inside a nested VM.
Firmware	OVMF (UEFI)	The image is UEFI-only; its protective MBR carries no boot code.
Privileges	none	No bridges, no tap devices, no root.

ADDRESSING

SEGMENT	SUBNET	XR1	XR2	OTHER END / NOTES
Loopback0	`/32`	`1.1.1.1`	`2.2.2.2`	Router identity; advertised into OSPF as passive, and the iBGP peering address
link1 — Gi0/0/0/0	`10.1.1.0/30`	`10.1.1.1`	`10.1.1.2`	Router-to-router
link2 — Gi0/0/0/1	`10.1.2.0/30`	`10.1.2.1`	`10.1.2.2`	Router-to-router
eBGP — Gi0/0/0/2	`10.2.1.0/30`	`10.2.1.1`	—	gobgp at `10.2.1.2`; xr1 only
OOB — MgmtEth0/RP0/CPU0/0	`10.99.0.0/24`	`10.99.0.11`	`10.99.0.12`	nms at `10.99.0.10`: syslog udp/514, SNMPv3 traps udp/162, and polls the routers on udp/161
Loopback98 (NaC-managed)	`/32`	`10.98.1.1`	`10.98.2.1`	Created by Terraform, not by day-0 config
Injected prefixes	`10.100.0.0/24 … 10.139.15.0/24`	10 000 × /24		Originated by gobgp; prefix *n* is `10.<100+n/256>.<n%256>.0/24`
OSPF router-id / area	`area 0`	`1.1.1.1`	`2.2.2.2`	Process `CORE`; both links point-to-point with MD5 key id 1
gNMI	`tcp/57400`	`no-tls`		How Terraform reaches the routers, tunnelled via nms

ROUTING PROTOCOLS

PROTOCOL	WHERE	DETAIL
OSPF `CORE`, area 0	both links + Loopback0 (passive)	Point-to-point. Each router learns the other's /32 over **both** links, giving two equal-cost paths — visible in CEF as a recursive load-share across Gi0/0/0/0 and Gi0/0/0/1. Both links require **MD5 authentication** (key id 1): no key, no adjacency.
BFD	2 single-hop + 1 multihop, per router	300 ms × 3 = 900 ms detection, against OSPF's 40 s dead interval. Single-hop on each link for OSPF; **multihop** between the loopbacks for iBGP, which needs `bfd multipath include location 0/0/CPU0` or it never leaves `BFD_MP_DOWNLOAD_NO_LC`. Not enabled towards gobgp — GoBGP has no BFD implementation.
iBGP AS 65000	Loopback0 ↔ Loopback0	Carries the 10 000 external prefixes from xr1 to xr2 with `next-hop-self`, so they arrive resolvable via `1.1.1.1`. Protected by multihop BFD.
eBGP AS 65000 ↔ 65100	xr1 Gi0/0/0/2 ↔ gobgp	gobgp originates 10 000 /24s. Inbound policy validates both the prefix range and the AS path before accepting any of them.
CDP	Gi0/0/0/0, Gi0/0/0/1	Each router sees exactly its peer on each link — the check that caught an early wiring bug where frames looped back to the sender.

ROUTING POLICY

IOS-XR applies **no default policy** to an eBGP neighbour: without one the session establishes and every prefix is silently discarded. Day-0 therefore installs a permissive `PASS` purely to bootstrap, and Network as Code replaces the inbound half with a policy that checks two independent things.

OBJECT	ON	PURPOSE
`NAC-GOBGP-PREFIXES`	xr1, xr2	prefix-set covering second octets 100-139 and nothing else — exactly the block gobgp generates. `ge 24 le 24` pins the length, so a more specific announcement inside the range does not match either.
`NAC-GOBGP-ASPATH`	xr1, xr2	as-path-set `ios-regex '^65100$'` — an AS path of exactly one hop, AS 65100: gobgp originated it and nothing else has touched it.
`NAC-GOBGP-IN`	xr1 (eBGP in)	Accepts only if the destination is in `NAC-GOBGP-PREFIXES` **and** the AS path is in `NAC-GOBGP-ASPATH`. Everything else is dropped.
`NAC-GOBGP-OUT`	xr1 (eBGP out)	`pass`.
`NAC-IBGP-IN`	xr2 (iBGP in)	Both checks applied again on what xr1 relays — defence in depth, since neither the prefix nor the AS path is rewritten inside an AS.

> **Why both checks:** an eBGP peer can advertise any prefix with any AS path it likes. A prefix in the right range carrying `65100 65200` — transited somewhere it should not have — is rejected by the AS-path set; a prefix with a perfect AS path but outside the block, or more specific than /24, is rejected by the prefix-set. The suite proves each independently, by originating prefixes that differ in only one of the two.

MANAGEMENT AND ACCESS

NODE	CONSOLE (TELNET)	SSH FROM HOST	QEMU MONITOR	CREDENTIALS
xr1	`127.0.0.1:5101`	`10.99.0.11 via nms`	`127.0.0.1:4101`	`admin`
xr2	`127.0.0.1:5111`	`10.99.0.12 via nms`	`127.0.0.1:4111`	`admin`
nms	`127.0.0.1:5121`	`127.0.0.1:2261`	`127.0.0.1:4121`	`lab`
gobgp	`127.0.0.1:5131`	`127.0.0.1:2262`	`127.0.0.1:4131`	`lab`

> **Only nms and gobgp have a host SSH forward.** XRv9k has a single management port, so it cannot sit on the OOB network *and* behind a NAT forward — the host has no route to the routers. `nms` bridges that: it is on the OOB network and reachable from the host, so it serves as the jump host for both the CLI and the test suite.

SNMPV3

ITEM	VALUE
Security level	`authPriv`
User / group	`labmon`
Auth / privacy	`SHA / AES-128`
View	`LABVIEW = 1.3.6.1 included`
Trap receiver	`10.99.0.10:162`
Trap source	`MgmtEth0/RP0/CPU0/0`
Traps enabled	`config · snmp linkup/linkdown · bgp cbgp2`
Engine IDs	`0x000000090300525400e90101 (xr1) · …e90201 (xr2)`

> **The engine ID is not the one you configure.** A v3 trap is authenticated against the sender's engine ID, and XRv9k accepts but ignores `snmp-server engineID local`, deriving it from the management MAC as `000000090300` + MAC. The receiver's `createUser` lines are keyed to those values; a mismatch drops the trap silently. There is no v1/v2c community anywhere in the lab.

VIRTUAL WIRING

SEGMENT	QEMU NETDEV	WHY
link1 / link2	`socket, unicast UDP pairs 11001↔11002, 11003↔11004`	A true point-to-point wire. Multicast sockets loop frames back to the sender, which made every router appear as its own CDP neighbour.
eBGP link	`socket, UDP 11005↔11006`	Same, to xr1's sixth vNIC.
OOB network	`socket, mcast 230.31.0.1:11099`	A shared segment for three nodes. Multicast loopback is harmless here: no link-layer neighbour protocol runs on it.
Host access	`user (SLIRP) + hostfwd`	nms and gobgp only.

vNIC order fixes XRv9k interface naming and cannot be varied: 1 MgmtEth0/RP0/CPU0/0 · 2 CtrlEth · 3 DevEth · 4 Gi0/0/0/0 · 5 Gi0/0/0/1 · 6 Gi0/0/0/2 (xr1 only). CtrlEth and DevEth are internal to a real chassis, but the image expects the vNICs, so they get dead-end hubs.

CONFIGURATION OWNERSHIP

LAYER	OWNS	DELIVERED BY
Day-0 `configs/<node>.cfg`	Bootstrap: hostname, credentials, management address, ssh, gRPC, bfd — plus the interfaces, OSPF (with MD5 auth and BFD) and iBGP the lab is built on	cvac, from a CD-ROM at first boot
Network as Code `nac/iosxr.nac.yaml`	Loopback98, the login banner, the AS-path and prefix sets, the route-policies, and which policies each BGP neighbour uses	Terraform over gNMI — `netascode/nac-iosxr` 0.1.1 / `CiscoDevNet/iosxr` 0.7.1

Day-0 must come first: it is what makes a router reachable by automation. Network as Code then reconciles what it owns — an out-of-band edit shows up as a `terraform plan` diff and is corrected by the next apply.

VERIFICATION

37 of 37 Robot Framework tests passing (about 2 min). Each run writes to its own timestamped directory under `test_results/`; sessions are tunnelled through nms, the same path an operator uses.

TAG	COVERS
`version, config`	Software release and that day-0 config actually applied
`cdp`	Each router sees exactly its peer on each link, on the matching remote port
`ospf, ecmp`	One FULL neighbour per link; the peer loopback learned over two equal-cost paths
`ospf, auth`	Message-digest authentication active on each link, key id 1
`bfd`	Both single-hop sessions and the multihop iBGP session up, with OSPF and BGP reporting them as clients
`bgp, ibgp`	Sessions established; peering uses the loopback addresses; all 10 000 prefixes relayed
`nms, oob, ssh, syslog`	OOB reachability, nms logging in to each router, syslog delivery per router
`snmp, traps, security`	authPriv polling of both routers, their traps arriving on nms, and a poll with the wrong password refused
`gobgp, prefixes`	All 10 000 originated, accepted and installed
`aspath`	A prefix with the wrong AS path is denied — end to end
`prefixset`	A prefix outside the permitted block, or more specific than /24, is denied — end to end
`nac`	NaC-managed config matches the model, and day-0 config survived the apply
`nac, banner`	The login banner matches the model, and is actually displayed when logging in